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The Holistic Health and Fitness Management System (H2FMS): a modernised software capability for optimising soldier readiness and performance

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ABSTRACT

The Holistic Health and Fitness Management System (H2FMS) is a secure, cloud-based platform designed to assess, understand and improve soldier health, fitness and performance across five readiness domains: physical, mental, spiritual, nutritional and sleep. By leveraging automated support actions, domain-specific tools, role-based dashboards and integrated analytics, H2FMS empowers the interdisciplinary Holistic Health and Fitness (H2F) Performance Teams (HPTs) to effectively develop, employ and retain talented soldiers within the US Army.

H2FMS unifies data from advanced biometric technologies, performance assessments, self-reported psychometrics and existing systems, ensuring every soldier has on-demand access to HPT support through guided education, interactive coaching, expert-informed training and personalised biofeedback. The platform enhances awareness for individual soldiers, HPT staff and cadre/command teams, driving optimal resource allocation and enabling data-informed interventions that boost performance, reduce injuries, expedite rehabilitation and significantly enhance quality of life and service. H2FMS represents a cost-efficient strategy for scaling HPT access across the Total Army, providing an equitable framework that maximises individual and organisational return on investment.

Critically, H2FMS aligns with the Army's modernisation and national defence objectives by supporting multi-domain overmatch through superior human performance management. It advances the long-term goal of generating and maintaining a lethal, adaptable and enduring force. This review details the requirements for H2FMS capabilities and illustrates how it can guide optimisation strategies while creating institutional efficiencies essential for modern force management. Ultimately, H2FMS marks a strategic milestone investment in modernising the foundation of the Army's 'human advantage,' ensuring sustained effectiveness and resilience in current and future conflicts.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Systems that integrate wearable technology, performance analytics and individualised training protocols are already in use by elite athletes and within some military organisations. The US Army's Holistic Health and Fitness interdisciplinary performance teams require a similar, modernised digital toolkit to maximise their effectiveness and efficiency by integrating advanced data-informed solutions to reduce injuries, behavioural health incidents and enhance combat readiness.

WHAT THIS STUDY ADDS

⇒ This paper highlights a comprehensive, Army-wide approach—combining cloud-based analytics, role-based dashboards and domain-specific training tools— that scales to meet the needs of a large and diverse force. By unifying holistic metrics, it expands on existing athlete management models and sets a new benchmark for integrated performance support in the military.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings and recommendations can guide future research on predictive analytics for injury prevention and readiness forecasting, while informing best practices for implementing large-scale digital health and fitness platforms. Policy-makers and leadership may leverage these insights to establish standardised, data-driven frameworks that streamline research allocation, enhance soldier care, and ultimately increase force lethality and resilience.

INTRODUCTION

Traditional methods of assessing and enhancing soldier readiness—primarily based on physical fitness tests and rudimentary medical screenings—are insufficient for modern warfare demands. Cognitive acuity, emotional resilience, nutrition, sufficient rest and well-being are now recognised as decisive factors in combat.[1–3] To address these complexities, the S Army formalised the Holistic Health and Fitness (H2F) System, a comprehensive approach at supporting five interdisciplinary domains of soldier readiness: physical, mental, spiritual, nutritional and sleep.[4 5] Staffed by a director, strength coaches, athletic trainers, occupational and physical therapy providers, nutrition experts, a cognitive performance specialist and a chaplain, unit-embedded H2F Performance Teams (HPTs) are central to the Army's performance optimisation strategy.[4–8] However, these professionals often manage large caseloads and competing requirements. Successfully implementing H2F at scale can overwhelm limited resources, especially without integrated digital solutions to streamline data management, automate administrative tasks and ensure equitable

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access to care.[9 10] Effectively and efficiently operationalising and quantifying H2F in the Total Army requires advanced data and technology systems capable of capturing, analysing and translating multidomain metrics into actionable insights.[5 11–13]

The Holistic Health and Fitness Management System (H2FMS) represents a strategic leap in delivering these capabilities. The platform's scalability and cost-effectiveness (less than 6% the cost of an HPT) provide blueprints for modernising the Army's human performance management strategy, aligning with broader modernisation, personnel care and readiness goals, while laying the foundation for multidomain overmatch.[4 5] By unifying biosensor data, personal assessments and training/educational content within an accessible framework, H2FMS aims to strengthen combat preparedness and reduce injuries, while maximising H2F's reach and return on investment.[11 14–16] As a software suite operating in a secure cloud environment, H2FMS offers a centralised platform for soldiers, HPTs and leaders to track, analyse and implement interventions for optimising performance and readiness.[9–11 17–19] In the Army's broader modernisation strategy, digital platforms like H2FMS aspire to harness the human 'data advantage,' where H2F domain information fuels descriptive, diagnostic, predictive and prescriptive analytics to convert raw inputs into actionable insights that guide interventions, refine training paradigms and forecast strategic requirements.[5 19–22] This powerful approach enhances the capacity to deliver timely, evidence-based support at every echelon.

Rationale

Modern combat often requires soldiers to operate under high physical and cognitive loads, in austere or technologically complex environments.[3–6 23 24] These operational realities drive emphasis on developing comprehensive health and performance solutions. Conventional fitness programmes, focused primarily on aerobic endurance and muscular strength, have been effective in building physical readiness. However, intangible factors—mental resilience, adequate sleep and spiritual well-being—are increasingly recognised as determinative of readiness.[1–6] H2FMS underpins this broader notion of 'holistic readiness,' providing an interconnected digital infrastructure for leaders and embedded experts to manage soldier resilience.

Declining fitness and rising obesity rates among young adults have shrunk the potential military recruiting pool.[2–6] Consequently, new accessions often arrive with suboptimal physical metrics and increased susceptibility to injury.[24 25] Simultaneously, mental health stressors among youth have risen, underscoring the necessity for advanced, integrative support from the onset of service.[26 27] H2FMS provides tools to track, intervene and coach individuals to meet readiness standards more quickly and sustainably.

An individual's decision to enlist, remain in service and maintain high-level performance depends on more than monetary compensation or patriotism. Quality-of-life factors, self-improvement opportunities and the perception that one's well-being is genuinely valued by an organisation are powerful motivators.[4 5] H2F, enabled by robust data systems like H2FMS, offers a structured approach to career-long development; this direct investment in each soldier yields double dividends: mission-ready personnel and improved recruitment/retention.

CAPABILITIES

H2FMS comprises two core capability areas: (1) a multiplatform software application and (2) a cloud environment for data aggregation. Combining real-time information capture, actionable data visualisations and advanced analytics, H2FMS supports HPTs implementing strategies for reducing injury risks, boosting performance and enhancing well-being. Figure 1 illustrates the H2FMS data ecosystem architecture.

Software

Role-based access

A role-based architecture is fundamental to H2FMS. New trainees, soldiers, HPTs and cadre/commands each require tailored interfaces and features, with specific data and functionalities necessary for their responsibilities. Automating 'least-privileged' permissions ensures protected health information (PHI) remains secure while relevant data are accessible for operational decisions. For instance, weekly HPT dashboards may highlight soldiers with elevated injury risks, while monthly readiness snapshots can inform Commanders.

Comprehensive training and education

H2FMS centralises holistic data to deliver 'intelligent training' recommendations and direct instruction. Instead of generic regimens/feedback, automated rules and analytics tailor interventions based on real-time performance indicators. After identifying overtraining or insufficient rest, H2FMS can automatically adjust workout intensity (or training load) and add targeted recovery protocols. 'Guided education' supports individualised progress through interactive modules on exercise technique, nutrition, mental skills, sleep optimisation and spiritual resilience. These modules are customisable, allowing mission-specific lessons, like climate adaptation or tactical stress management.

Biotechnology integration and biofeedback

Biometric devices, like wearables, capture physiological and biomechanical measures of workload and recovery.[14 15 20] Metrics like heart-rate variability, step counts and sleep quantity/quality integrate from wearable devices, drawing near real-time data streams into a single analytical environment. When wearables are integrated, H2FMS pushes feedback to soldiers and HPTs, prompting timely awareness to anticipate injury or fatigue risks and direct swift interventions.[9–12] For example, soldiers or leaders might receive immediate alerts about potential dehydration or excessive mental strain, with automated guidance on mitigation.

Digital assessments

Performance evaluations, like the Army Combat Fitness Test and rifle marksmanship, historically operate in silos. H2FMS consolidates recording and analysing these assessments into a single platform, correlating outcomes with relevant domain data. Leaders can quickly identify patterns and take pre-emptive steps before issues escalate. These combined datasets yield insights for refining training schedules, guiding resource allocation and evaluating the effectiveness of policies aimed at maximising performance.[19 22]

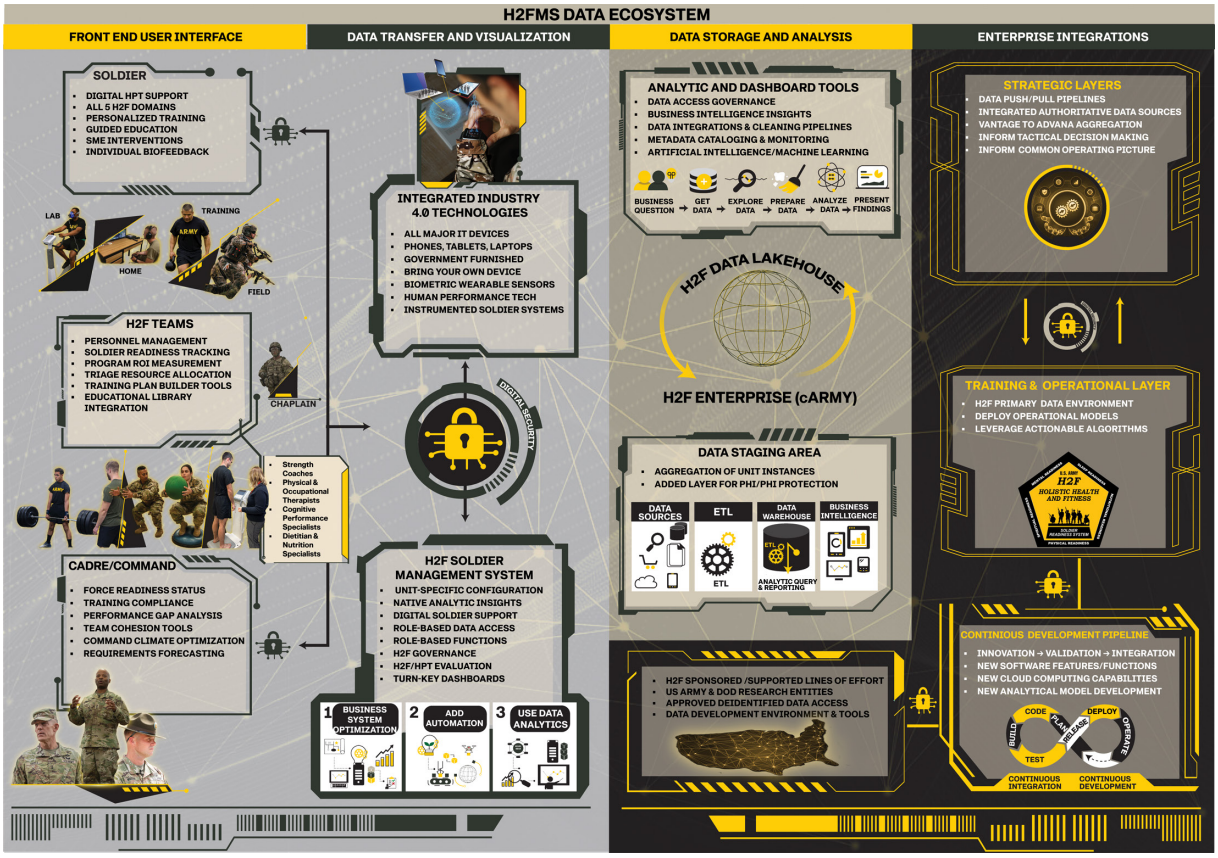


Figure 1 H2FMS data ecosystem architecture. The appearance of US Department of Defense (DoD) visual information does not imply or constitute DoD endorsement. This diagram illustrates the comprehensive structure of H2FMS, highlighting the integration of various data sources (eg, wearable devices, performance assessments, surveys and existing Army systems) into a secure cloud infrastructure. H2FMS processes and analyses data through an integrated analytics engine and data processing modules, delivering personalised training plans, guided education modules, interactive coaching, real-time biofeedback and automated alerts via role-based dashboards. The soldier's dashboard features personal performance metrics, training plans, biofeedback and wellness tips, enabling self-monitoring and personalised goal setting. The H2F Performance Team's (HPTs) dashboard consolidates soldier data, highlights injury risk indicators and provides intervention tools and resource management features, facilitating targeted support and efficient resource allocation. The Drill Sergeants' dashboard offers unit readiness snapshots, training progress reports and soldier performance summaries, aiding in monitoring unit readiness and training compliance. Finally, the Commanders' dashboard presents aggregated readiness data, strategic resource allocation tools and high-level performance indicators, supporting strategic decision-making and overall force readiness assessment. These role-based dashboards ensure that each user accesses relevant and actionable insights tailored to their specific responsibilities, enhancing decision-making and operational efficiency across the Army hierarchy. This architecture ensures seamless data flow and supports the holistic management of soldier health, fitness and performance across multiple readiness domains. H2FMS, Holistic Health and Fitness Management System.

Surveys and scheduling

Beyond objective data, soldiers periodically complete surveys capturing subjective well-being, motivation or stress.[5] These insights enrich the analytical engine, providing nuanced views of potential challenges, responses and adaptations. Where surveys reveal mounting stress, H2FMS can automatically notify leaders or schedule pastoral/behavioural-health appointments, reducing HPT manual coordination workloads.[9–12 27] Scheduling capabilities extend beyond individual sessions. Leaders can reserve gym time, coordinate group training events or manage facility usage. Automated notifications keep soldiers informed of upcoming tasks, minimising confusion and maximising availability. Through centralised functionality, H2FMS streamlines administration, ensuring soldiers and HPTs focus on high-value tasks.

Automated reporting and coaching interface

Manual data entry and time-consuming documentation historically burden human performance staff. By streamlining data capture and providing user-friendly, role-based dashboards with triaged information, H2FMS frees up time for more direct soldier engagement. Features like 'click-to-build' training routines democratise best practices: any soldier can rapidly assemble training plans meeting H2F standards. Crucially, many support actions are automated. When data surpasses predetermined thresholds (eg, sleep duration consistently falls below acceptable levels), H2FMS can autogenerate recommendations or schedule time with specialists, reducing administrative load by eliminating countless hours spent manually tracking and following up with soldiers.[3 10–12 16] Table 1 summarises H2FMS software performance requirements, target states and evaluation metrics. Figure 2 identifies a few H2FMS applications across H2F readiness domains.

Table 1 identifies the H2FMS software performance requirements, target states for operational integration and utilisation performance metrics for continuous capability analysis and refinement.

Table 1 H2FMS software performance requirements

Capability area 1. Software application to support and facilitate H2F across the total army		
Performance measure	**Target state**	**Metrics to consider**
Single software application for all 5 H2F readiness domains	95% of H2F staff and end users use a single software platform for training, education and access to care in all H2F domains	User adoption rate, reach, platform usage and impact metrics, user satisfaction surveys, manpower hours saved
Role-based configuration to control H2F data access and software function	99% of end users have role-based access controls with least privileged permissions, with 95% of data access requests approved within 24 hours	Access control audit logs, user role configuration metrics, security incident reports
Access to H2F Performance Teams	90% of end users have digital access to H2F subject matter experts, with 80% of users reporting satisfactory support	User access metrics, latency, support request metrics, user satisfaction surveys, impacts on readiness
H2F personnel management	95% of H2F HPT staff activities and encounters are tracked and quantified, with 90% of scheduling and calendar integration automated	Data accuracy metrics, data completeness metrics, user satisfaction surveys
H2F training plans and tools for all 5 readiness domains	85% of training plans are developed using guided tools, with 80% of users reporting improved training outcomes	Training plan adoption rate, training plan completion rate, user satisfaction surveys, impacts on readiness
H2F education plans and tools for all 5 readiness domains	85% of education plans are developed using guided tools, with 80% of users reporting improved education outcomes	Education plan adoption rate, education plan. completion rate, user satisfaction surveys, check on learning performance
H2F facilities management	90% of H2F facilities and activities are tracked and quantified, with 85% of scheduling and calendar integration automated	Facility utilisation metrics, scheduling accuracy metrics, user satisfaction surveys
H2F software monitoring	99% of software performance and utilisation data is tracked and reported, with 95% of issues resolved within 24 hours	Software uptime metrics, issue resolution time metrics, user satisfaction surveys
H2F data ingestion for all 5 readiness domains	90% of data are ingested through automated pathways, with 85% of data ingested within 24 hours	Data ingestion metrics, data processing metrics, data analytics metrics
H2F data presentation	90% of users can access and use actionable data insights through multiple presentations methods to inform H2F decisions	User adoption rate, data presentation metrics, user satisfaction surveys
H2F business rule automation	85% of business rules are automated, with a 95% success rate in executing software functional operations based on specified data inputs	Business rule automation metrics, decision-making metrics, user interaction metrics, user satisfaction surveys, manpower hours saved, reach and access metrics, impacts on readiness
H2F data lakehouse integration	95% of H2F software instances are integrated into a single, secure government-owned cloud environment, with 99% data availability and 95% data integrity	Data integration metrics, data availability metrics, data integrity metrics

H2FMS, Holistic Health and Fitness Management System; HPT, H2F Performance Team.

Cloud

Data aggregation and analytics

The system's cloud-based data repository facilitates large-scale analytics, including machine learning and artificial intelligence models for predicting injury risks and performance trends.[18–21] Enterprise-wide implementation supports model development for optimisation strategy selection across H2F domains. In practice, soldiers choose from prescriptive training/education plans modelled to their individual requirements, while commanders make unit training modifications based on predictive risk insights. The Army at scale would be informed by the richest, most advanced human data advantage in history.

Systems interoperability

To bolster tactical and strategic situational awareness, H2FMS seamlessly integrates with Army systems of record and operational data sources. H2FMS pushes and pulls essential readiness data with these systems, bridging silos that previously impeded holistic care.[1–3 12] This interoperability reduces redundant data entry, ensures record consistency and enriches H2FMS analytics with contextual information, ultimately informing more robust tactical decisions and long-range planning.[19–21]

Continuous innovation pathway

To future-proof H2FMS, the cloud infrastructure includes a research and development-focused partition—de-identified environment for scientific and technological innovation. Approved research labs and collaborating universities can develop and validate new analytical models or features/functions, furthering best practices in soldier health, performance and injury prevention. This approach replicates proven 'living laboratory' models, where continuous user feedback and large-scale data streams accelerate discovery.[16] By maintaining this research-to-implementation continuous loop, H2FMS ensures H2F remains at the cutting edge of evidence-based solutions, enabling the Army to iteratively refine protocols and integrate emerging insights.[13 18–22] Table 2 summarises H2FMS cloud performance requirements, target states and evaluation metrics.

Table 2 identifies the H2FMS cloud performance requirements, target states for operational integration and utilisation performance metrics for continuous capability analysis and refinement.

ARMY-WIDE APPLICATIONS

H2FMS integrates with Army structures and interacts according to end-user needs across varying roles, career

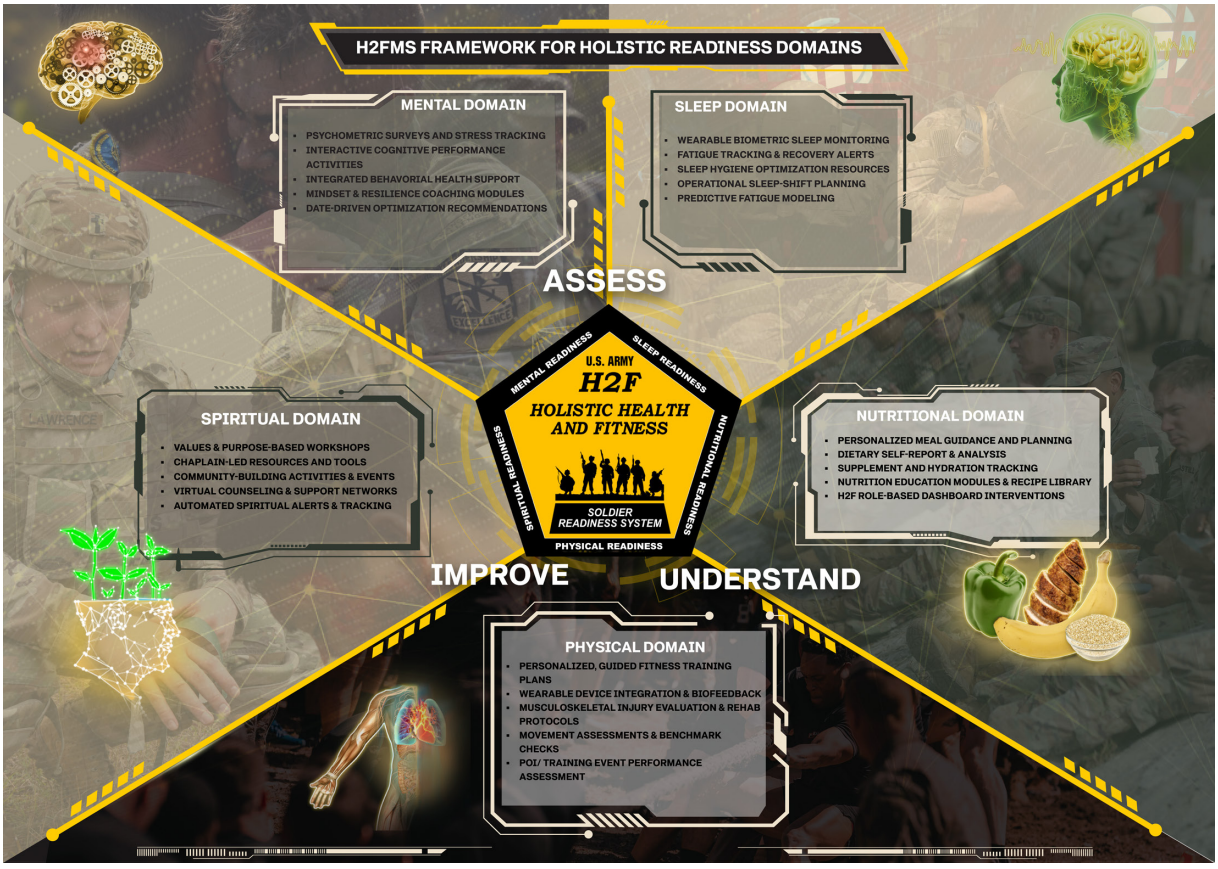


Figure 2 H2FMS capabilities across the five H2F readiness domains. The appearance of US Department of Defense (DoD) visual information does not imply or constitute DoD endorsement. This figure illustrates some of H2FMS's integrated tools, features and functions that support soldiers' readiness in the physical, mental, spiritual, nutritional and sleep domains. In the physical domain, soldiers and H2F Performance Team (HPT) staff use wearable devices, movement assessments and automated workout prescriptions to optimise strength, endurance and injury prevention. For mental resilience, the system offers stress assessments, mindfulness modules and early-warning alerts that empower both individuals and HPT professionals to intervene proactively. The spiritual domain is supported through values-based coaching, access to chaplain-led resources and tailored programmes fostering a sense of purpose and community. In the nutritional domain, soldiers can track dietary habits, receive personalised meal recommendations and schedule consultations with dietitians—all coordinated via a unified H2F dashboard. Finally, the sleep domain leverages biometric and self-reported data to monitor sleep quality, trigger rest and recovery recommendations, and automate appointments with specialists when chronic issues arise. Together, these domain-specific capabilities feed into a role-based analytics engine that helps HPTs and command staff identify at-risk individuals or units early, implement targeted interventions and adjust training regimens as needed. By consolidating biometric data, performance metrics, guided training and interactive educational content in one secure cloud platform, H2FMS creates a 360° view of soldier readiness—facilitating evidence-based decisions and supporting a healthier, more resilient force across all echelons. H2FMS, Holistic Health and Fitness Management System.

Table 2 H2FMS cloud performance requirements

Capability area 2. GovCloud environment to aggregate, integrate and apply H2F data at scale		
Performance measure	**Target state**	**Metrics to consider**
Data storage	99% of all H2F-related data are stored in a secure, scalable, and accessible data lakehouse, with 95% of data available for analytics within 24 hours.	Data storage capacity metrics, data retrieval metrics, data security metrics
Cloud usage tracking	95% of cloud usage is accurately tracked and quantified, with 90% of costs allocated to the correct billpayer	Cloud usage metrics, cost allocation metrics, billing accuracy metrics
Systems integration	90% of scheduled and triggered data pushes/pulls with other Army and DoD data systems of record are successful, with 85% of data exchanges occurring within 24 hours.	Integration success metrics, data exchange metrics, system downtime metrics
Unit data analytics	85% of units have access to tools for performing statistical analyses and AI/ML operations, with 80% of users reporting improved data-driven decision-making	User adoption metrics, analytics usage metrics, survey decision-making metrics
Aggregate data analytics	90% of echelons have access to tools for performing statistical analyses and AI/ML operations, with 85% of users reporting improved data-driven decision-making	User adoption metrics, analytics usage metrics, survey decision-making metrics
Deidentified S&T partition	95% of deidentified data is made available to authorised Army organisations and entities, with 90% of users reporting satisfactory access and utilisation	Data access and governance metrics, user satisfaction surveys, innovation metrics

AI, artificial intelligence; DoD, Department of Defense; H2FMS, Holistic Health and Fitness Management System; ML, machine learning; S&L, science and technology.



Figure 3 H2FMS applications across a soldier's career. The appearance of US Department of Defense (DoD) visual information does not imply or constitute DoD endorsement. This graphic illustrates the end-to-end integration of H2FMS software and cloud capabilities throughout a soldier's entire career. Beginning at the recruiting stage, the system provides prospective recruits with upfront training and education resources—delivered through secure cloud access and user-friendly software interfaces—to build physical and mental resilience before day 1 of basic training. This early, data-driven engagement reduces injuries and attrition, promotes socialisation and unit cohesion and helps improve accession outcomes by equipping new soldiers with foundational knowledge and support. As soldiers transition into training and doctrine command environments, H2FMS adapts to their evolving needs, providing continuous education, skill development and performance tracking. Real-time analytics and predictive modelling support individualised training regimens that optimise readiness while minimising the risk of injuries. Forces command then leverages H2FMS for combat readiness, seamlessly integrating with predeployment screenings and postdeployment recovery protocols. This ensures holistic monitoring of physical, mental, spiritual, nutritional and sleep domains, ultimately enhancing resilience and operational capability. For the reserve and national guard, who often serve in geographically dispersed units and have reduced duty hours, the H2FMS cloud-based platform allows for continuous, flexible access to critical performance tools and expert consultations, regardless of location. Finally, as service members transition to the Department of Veterans Affairs, H2FMS provides continuity of care by securely transferring relevant health and performance data, supporting ongoing wellness and maximising cost-effectiveness for both the soldier and the healthcare system. At each step, data generated by H2FMS is automatically aggregated within the secure cloud and interoperable with other systems of record. This aggregated, longitudinal dataset fuels artificial intelligence (AI) and machine learning (ML) models, enabling predictive and prescriptive insights that inform tactical decisions on training and readiness, as well as strategic planning for force management. In doing so, H2FMS delivers a unified, data-driven 'common operating picture' that drives smarter resource allocation, personalised support and stronger outcomes for soldiers' entire career lifecycle and beyond to continuously 'deliver the human advantage'. H2FMS, Holistic Health and Fitness Management System.

progressions and available resources. Figure 3 illustrates how H2FMS can be implemented by every soldier throughout the Army.

Recruits and trainees

On entry to service, many face steep learning curves managing fitness, nutrition and stress.[3–6] Trainees frequently struggle with injuries and the psychological demands of their first military experience.[24 25] H2FMS administers holistic baseline assessments to unify data and generate individualised training/education prescriptions for development prior to basic training. By connecting recruits with their supporting HPT, the Army can accelerate learning curves for military skills, increase accession rates, incoming readiness and decrease the likelihood of injury or attrition.

Soldiers

H2FMS serves as a personal readiness companion, blending interactive coaching with guided optimisation. Soldiers can access personal dashboards for objective self-monitoring, tracking progress and receiving critical, tailored support from HPTs. H2FMS offers individualised and guided training/education plans across all five H2F domains, recommended recovery strategies, the ability to schedule practitioner evaluations and automated alerts to drive healthy

behaviour decisions (eg, hydration prompts during environmental conditions or mindfulness exercises after elevated stress). This proactive loop helps sustain mental and physical resilience, particularly under high operational tempo.

H2F Performance Teams

Unit-embedded human performance experts are the core of H2F's success. H2FMS alleviates administrative strain by integrating data streams and automating routine support actions (eg, auto-booking follow-up visits if pain fails to improve or scheduling group nutrition classes when unit body composition exceeds threshold). H2FMS can further optimise workflows and maximise HPT return on investment by highlighting at-risk soldiers and sharing data-informed best practices to guide interventions.[9–12] By minimising tedious manual processes, HPTs can focus their expertise where it matters most. Instead of deciphering spreadsheets, consolidated dashboards illuminate patterns of poor recovery, high stress or nutritional gaps.[19–22] With these analytical insights, HPTs can deliver more targeted support at scale, augmenting impact while conserving manpower.

Command teams

Cadre and unit leaders require aggregated readiness snapshots to inform training cycle decisions. By weaving group metrics into a 'common operational picture,' H2FMS helps leaders allocate resources (eg, adjusting march intensities, scheduling rest days, requesting support for high-risk subsets). Embedded predictive analytics provide early warnings of rising stress or potential mental health concerns, ensuring timely allocation of resources.[18 19 27] Small-unit leaders leverage aggregated metrics to optimise training. Instead of anecdotal evidence, they have data-driven insights on who needs rest, additional conditioning or mental-skills training. Automated alerts and behind-the-scenes processes reduce the burden of combing through raw data, allowing leaders to concentrate on command tasks, effectively compressing the time between identifying readiness gaps and implementing corrective measures.[20 21] This supports broader strategies of using reliable, timely information to maintain a lethal, balanced force.

Senior leadership

At higher echelons, H2FMS informs strategic decision-making. Over time, leadership may observe how sleep and recovery shortfalls or mental strain correlate with retention issues. These insights underpin decisions about budget allocation, facility upgrades, hiring specialised staff or revising doctrine. By providing an accurate, holistic picture of readiness, H2FMS exemplifies the commitment to data-driven modernisation and multidomain overmatch. These data advantages bolster near-term readiness and the capacity to anticipate future operational demands or challenges.[1–5 10–13]

DELIVERING THE HUMAN ADVANTAGE

Investing in holistic, modernised solutions like H2FMS confers strategic and economic benefits. The system drives down healthcare costs and retraining expenses by identifying potential risks early. Automation further frees HPTs to work their skillsets improving soldier outcomes rather than managing paperwork. This cost-efficiency ensures every soldier, regardless of unit or location, receives the support they deserve. Longer-term, equipping soldiers and units with continuous feedback loops and robust performance analytics strengthens national defensive posture. As operational landscapes evolve, H2FMS will adapt to new data sources and demands, ensuring the Army remains agile and effective. Thus, the system's economic and strategic value lies not only in near-term injury risk reduction but also in forging a resilient, career-long path to lifelong optimisation.

Holistic readiness translates to increased lethality. Empirical research confirms physically fit, mentally resilient units outperform less prepared peers in high-stress operational scenarios.[3 6 24] By providing a single platform for holistic readiness and performance, H2FMS empowers immediate interventions to ensure warfighters are mission-capable when it matters most. Soldier morale may be heightened when they perceive leadership invests in comprehensive wellness strategies. H2FMS's capacity for personalised alerts enables soldiers to self-regulate and better manage personal wellness. Microlevel interventions (eg, customised dietary tips, mindfulness exercises) can yield macrolevel benefits, including lower attrition rates and stronger unit cohesion.

Musculoskeletal injury, mental health and lost productivity cost the Army billions every year.[4–7 23–26] By integrating analytics with real-time data, H2FMS flags rising injury risks—inadequate sleep, excessive training load or behavioural concerns—long before costly interventions become necessary. Even a modest 10% drop in injuries can yield over US$500 million in annual savings across direct healthcare, lost productivity and retention budgets.[26] 1 year of returns could fund H2FMS for nearly a decade. By streamlining scheduling and generating individualised training or rehabilitation protocols, the platform supports early, targeted interventions that help soldiers avoid debilitating conditions and potential medical separation. This proactive approach elevates morale, fosters resilience and sustains a healthier, more productive force. In turn, increased retention and readiness ultimately strengthen the Army's long-term human capital and strategic effectiveness.

In an era of near-peer competition, maintaining a healthy, agile force is critical to national defence. The Army People Strategy posits that readiness depends on acquiring, developing, employing and retaining talented soldiers.[4] By offering a robust, data-enabled environment for fostering growth, H2FMS reinforces the Army's attractiveness to modern recruits, who often favour technology-driven, personalised solutions.[9 10] The net effect is a more stable, skilled and engaged future force. With H2FMS, decision-makers can incorporate near real-time readiness data into operational plans, ensuring the 'human advantage' remains a decisive factor on tomorrow's battlefield.[5 18–22] Collectively, these alignments reinforce the ongoing transition to a technology-integrated, people-focused and readiness-centric institution, better positioned to achieve multidomain overmatch for the foreseeable future.

Security

H2FMS operates with personally identifiable information and PHI, necessitating strict controls on data handling, storage and access to prevent morale-damaging breaches. The system must meet at least impact level IV standards, requiring encryption, intrusion detection and role-specific authentication.[28] H2FMS must incorporate continuous monitoring and penetration testing to ensure data integrity, trust and compliance.

Culture

Implementing H2FMS requires more than deploying technology—it demands an organisational mindset valuing trust, ownership and transparency. Soldiers must understand how

their data leads to tangible benefits and feel assured their information remains secure. Clear, proactive communication from leaders fosters acceptance, ensuring H2FMS is seen as empowering rather than punitive. When soldiers feel ownership of their data and see personal advantages, they are more likely to engage, report honestly and follow recommendations.

Doctrine, Organisation, Training, Materiel, Leadership & Education, Personnel, Facilities, Policy (DOTmLPF-P)

H2FMS must be sustainably woven into the fabric of day-to-day operations. Doctrine must standardise and disseminate best practices. Organisational responsibilities for unit integration should be defined. Regular utilisation in training throughout a career would maximise benefit and proficiency. Standardising requirements and interoperability will ensure material solutions remain resilient and avoid redundancy. Leader education could promote the proactive, data-driven management of holistic unit readiness. A cross-functional H2FMS team will be essential to integrating and sustaining the capability across the force. Future soldiers with H2F occupational specialties could serve as critical action officers. Many current facilities require updated infrastructure to embrace connectivity with emerging technologies. Clear policies must define who can access data, for what purposes and under which circumstances. Orders and regulations should task and enforce comprehensive integration, while a long-term, dedicated budget line will allow H2FMS to deliver its promise of advanced data-driven care, cost savings, injury prevention and overall force readiness.

CONCLUSIONS

Anchored in delivering future soldiers with decisive human advantage, H2FMS marks a pivotal turning point for the US Army's holistic approach to readiness. The impetus behind H2FMS is the Army's vision of the 'future soldier': seamlessly integrating advanced personal analytics, mission command systems and domain-specific training to achieve overmatch against near-peer adversaries.[3–5] As emerging threats demand greater agility, mental acuity and holistic well-being, the synergy of software, analytics and embedded HPTs represents a decisive edge. H2FMS stands at the nexus of technology, human performance and military strategy: an integrated platform tailored to the holistic domains of readiness and developing lethal soldiers. Its capacity to unify data streams, deliver timely interventions and streamline resource allocation ensures every soldier can access crucial performance support. In doing so, the Army enhances combat efficiency and invests in sustainable, long-term force readiness.

While concerns over privacy and cultural acceptance must be addressed, the long-term potential benefits—in cost savings, injury reduction, heightened readiness—are clear. H2FMS offers a forward-looking solution for preserving, protecting and enhancing the human dimension of combat power. More than a technological solution, it is a strategic enabler of the broader shift toward holistic, data-driven soldier readiness. H2FMS pays for itself and provides an economically efficient solution for scaling HPTs to the Army, ensuring every soldier benefits from high-quality, data-informed support. By delivering consistent feedback loops across a soldier's career, the system facilitates lifelong development, talent retention and a more resilient force. As modern conflict demands evolve, so must the means to prepare and sustain those who serve. Thus, H2FMS represents a critical milestone in the Army's journey toward future-focused readiness and the essential human advantage in 21st-century warfighting.

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